

```

1 import pygame, sys
2
3 # Define the screen size
4 SCREEN_WIDTH = 800
5 SCREEN_HEIGHT = 600
6
7 SCREEN_SIZE = (SCREEN_WIDTH, SCREEN_HEIGHT)
8
9 # Initialize Pygame
10 pygame.init()
11 pygame.font.init()
12
13
14 class Checkbox:
15     def __init__(self, surface, x, y, idnum, color=(
16         230, 230, 230),
17         caption="", outline_color=(0, 0, 0
18         ), check_color=(0, 0, 0),
19         font_size=22, font_color=(0, 0, 0),
20         text_offset=(28, 1), font='Ariel
21         Black'):
22         self.surface = surface
23         self.x = x
24         self.y = y
25         self.color = color
26         self.caption = caption
27         self.oc = outline_color
28         self.cc = check_color
29         self.fs = font_size
30         self.fc = font_color
31         self.to = text_offset
32         self.ft = font
33
34         # identification for removal and
35         reorginazation
36         self.idnum = idnum
37
38         # checkbox object
39         self.checkbox_obj = pygame.Rect(self.x, self.
40         y, 12, 12)
41         self.checkbox_outline = self.checkbox_obj.
42         copy()
43
44         # variables to test the different states of

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38 the checkbox
39     self.checked = False
40
41     # total keeps track of the total price
42     #self.total = Prices[idnum + 1] if idnum in
Prices else 0
43     self.total = 0
44
45     def _draw_button_text(self):
46         self.font = pygame.font.SysFont(self.ft, self
.fs)
47         self.font_surf = self.font.render(self.
caption, True, self.fc)
48         w, h = self.font.size(self.caption)
49         self.font_pos = (self.x + self.to[0], self.y
+ 12 / 2 - h / 2 +
50                             self.to[1])
51         self.surface.blit(self.font_surf, self.
font_pos)
52
53     def render_checkbox(self):
54         if self.checked:
55             pygame.draw.rect(self.surface, self.color
, self.checkbox_obj)
56             pygame.draw.rect(self.surface, self.oc,
self.checkbox_outline, 1)
57             pygame.draw.circle(self.surface, self.cc
, (self.x + 6, self.y + 6), 4)
58
59         elif not self.checked:
60             pygame.draw.rect(self.surface, self.color
, self.checkbox_obj)
61             pygame.draw.rect(self.surface, self.oc,
self.checkbox_outline, 1)
62             self._draw_button_text()
63
64     def _update(self, event_object):
65         x, y = pygame.mouse.get_pos()
66         px, py, w, h = self.checkbox_obj
67         if px < x < px + w and py < y < py + w:
68             if self.checked:
69                 self.checked = False
70                 self.total -= Prices.get(self.idnum
+ 1, 0) # subtract prices if unchecked

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71         else:
72             self.checked = True
73             self.total += Prices.get(self.idnum
+ 1, 0) # add price if its checked
74
75     def update_checkbox(self, event_object):
76         '''checked = [False, False, False, False,
False, False, False, False]
77         total = 10 # initial price of Pizza
78
79         checked = [box.checked for box in boxes] #
checking if each box is True or False
80
81         Prices = {1: 1.5, 2: 1, 3: 2.5, 4: 1.5, 5:
1.5, 6: 0.5, 7: 0.25, 8: 3}
82
83         for i in range(8):
84             if checked[i] == True: #making sure to
include initial price of the pizze as $$10
85                 topping = Prices.get(i+1)
86                 total += topping'''
87
88         if event_object.type == pygame.
MOUSEBUTTONDOWN:
89             self.click = True
90             self._update(event_object)
91
92
93 # Create the screen
94 screen = pygame.display.set_mode((SCREEN_WIDTH,
SCREEN_HEIGHT))
95
96 # Create two surfaces to split the frame
97 surface1 = pygame.Surface((SCREEN_WIDTH * 0.3,
SCREEN_HEIGHT))
98 surface2 = pygame.Surface((SCREEN_WIDTH * 0.7,
SCREEN_HEIGHT))
99
100 # Fill the surfaces with different colors
101 surface1.fill((255, 255, 255))
102 surface2.fill((0, 200, 0))
103
104 # Blit the surfaces to the screen
105 screen.blit(surface1, (0, 0))

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106 screen.blit(surface2, (SCREEN_WIDTH * 0.3, 0))
107
108 boxes = []
109
110 Prices = {1: 1.5, 2: 1, 3: 2.5, 4: 1.5, 5: 1.5, 6: 0
    .5, 7: 0.25, 8: 3}
111
112 button1 = Checkbox(screen, 50, 200, 0, caption='
    Pepparoni $1.50')
113 button2 = Checkbox(screen, 50, 225, 1, caption='
    Mushrooms $1.00')
114 button3 = Checkbox(screen, 50, 250, 2, caption='
    Onions $2.50')
115 button4 = Checkbox(screen, 50, 275, 3, caption='
    Bell Peppers $1.50')
116 button5 = Checkbox(screen, 50, 300, 4, caption='
    Olives $1.50')
117 button6 = Checkbox(screen, 50, 325, 5, caption='
    Basil $0.50')
118 button7 = Checkbox(screen, 50, 350, 6, caption='
    Oregano $0.25')
119 button8 = Checkbox(screen, 50, 375, 7, caption='
    Garlic $3.00')
120
121 boxes.append(button1)
122 boxes.append(button2)
123 boxes.append(button3)
124 boxes.append(button4)
125 boxes.append(button5)
126 boxes.append(button6)
127 boxes.append(button7)
128 boxes.append(button8)
129
130 #https://stackoverflow.com/questions/68331123/pygame
    -checkbox-class-works-locally-but-not-when-imported
131 #https://pyga.me/docs/tutorials/en/intro-to-pygame.
    html
132
133 #I did the above code
134 #my partner did the below code
135
136 circleX = 500
137 circleY = 300
138 radius = 150

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139 Khaki = ("Khaki1")
140
141 radius2 = 130
142 Red = ('Red')
143
144 # Update the display
145 pygame.display.flip()
146
147 baseprice = 10.00
148
149 # Run the game loop
150 while True:
151
152     # Check for events
153     for event in pygame.event.get():
154         # If the user clicks the any of the
checkboxes, highlight that checkboxe
155         if event.type == pygame.MOUSEBUTTONDOWN:
156             for box in boxes:
157                 box.update_checkbox(event)
158
159         # If the user clicks the close button, quit
the game
160         if event.type == pygame.QUIT:
161             pygame.quit()
162             sys.exit()
163
164     # total price
165     total = baseprice #used ChatGPT for logic
166     for box in boxes:
167         total+= box.total
168
169
170     # Draw the frame
171     screen.fill((0, 0, 0))
172     screen.blit(surface1, (0, 0))
173     screen.blit(surface2, (SCREEN_WIDTH * 0.3, 0))
174
175     for box in boxes:
176         box.render_checkbox()
177
178     pygame.draw.circle(screen, Khaki, (circleX,
circleY), radius) # DRAW CIRCLE
179     # Load the background image

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```

180     Pepperoni = pygame.image.load("Pepperoni.png")
181     Mushrooms = pygame.image.load("Mushrooms.png")
182     Onion = pygame.image.load("Onion.png")
183     Bell_Peppers = pygame.image.load("Bell Peppers.
png")
184     Olives = pygame.image.load("Olives.png")
185     Basil = pygame.image.load("Basil.png")
186     Oregano = pygame.image.load("Oregano.png")
187     Garlic = pygame.image.load("Garlic.png")
188
189
190     #resize the images to fit
191     Pepperoni = pygame.transform.scale(Pepperoni, (
50, 50))
192     Mushrooms = pygame.transform.scale(Mushrooms, (
50, 50))
193     Onion = pygame.transform.scale(Onion, (50,50))
194     Bell_Peppers = pygame.transform.scale(
Bell_Peppers, (50,50))
195     Olives = pygame.transform.scale(Olives, (50,50))
196     Basil = pygame.transform.scale(Basil, (50,50))
197     Oregano = pygame.transform.scale(Oregano, (50,50
))
198     Garlic = pygame.transform.scale(Garlic, (50,50))
199
200
201     # Blit the background image to the screen
202     #screen.blit(background_image, (0, 0))
203     pygame.draw.circle(screen, Red, (circleX,
circleY), radius2)
204
205     font = pygame.font.SysFont('Arial', 25)
206     font.set_bold(True)
207     pygame.draw.rect(screen, "red", (50, 425, 165,
50))
208     # have to display total now
209     text_surface = font.render('Total : $' + str(
total), True, 'Black') # Generated by ChatGPT for
syntax
210     screen.blit(text_surface, (70, 435))
211
212     if button1.checked:
213         screen.blit(Pepperoni, (circleX - 50,
circleY - 50))

```

```
214
215     if button2.checked:
216         screen.blit(Mushrooms, (circleX + 0, circleY
217     - 50))
218
219     if button3.checked:
220         screen.blit(Onion, (circleX - 100, circleY
221     - 50))
222
223     if button4.checked:
224         screen.blit(Bell_Peppers, (circleX - 50,
225     circleY - 0))
226
227     if button5.checked:
228         screen.blit(Olives, (circleX - 50, circleY
229     - 100))
230
231     if button6.checked:
232         screen.blit(Basil, (circleX - 0, circleY - 0
233     ))
234
235     if button7.checked:
236         screen.blit(Oregano, (circleX - 100, circleY
237     - 0))
238
239     if button8.checked:
240         screen.blit(Garlic, (circleX - 0, circleY -
241     100))
242
243     # Update the display
244     pygame.display.flip()
245
246
247
```